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Microglial Activation Visualized by [^{18}F]-DPA714 PET Is a Potential Marker of Severity and Prognosis for Anti-LGI1 Encephalitis

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Keywords: anti-leucine-rich glioma-inactivated 1protein encephalitis | biomarker | microglia | prognosis | translocator protein

ABSTRACT

Background and Purpose: Whether microglial activation plays an important role in the pathogenesis of autoimmune encephalitis (AE), such as anti-leucine-rich, glioma-inactivated-1 (LGI1) encephalitis, remains unknown. [^{18}F]-DPA714 PET targeting the translocator protein (TSPO) is a novel method to detect neuroinflammation via visualizing activated microglia. In this study, we aimed to investigate the application of [^{18}F]-DPA714 PET in anti-LGI1 encephalitis.

Methods: Patients with anti-LGI1 encephalitis and non-inflammatory controls (NIC) underwent [^{18}F]-DPA714 PET scans were enrolled. Standardized uptake value ratios normalized to the cerebellum (SUVRc) in LGI1-AE patients were calculated for semi-quantitative analysis. The microglial activation marker, soluble triggering receptor expressed on myeloid cells 2 (sTREM2) was measured in cerebrospinal fluid (CSF) to demonstrate its correlation with [^{18}F]-DPA714 PET imaging. Logistic regression analysis was used to identify potential predictors of prognosis.

Results: Forty-six patients with anti-LGI1 encephalitis were included in this study. Increased TSPO uptake was identified in the hippocampus, frontal cortex, and caudate nucleus. Montreal Cognitive Assessment (MoCA) score was significantly correlated with SUVRc in the hippocampus ($R^2 = 0.13$, $p = 0.034$) and frontal cortex ($R^2 = 0.13$, $p = 0.017$). Overexpressed sTREM2 in CSF was correlated with SUVRc in the hippocampus ($R^2 = 0.18$, $p = 0.04$). SUVRc in the hippocampus significantly decreased after immunotherapy and was associated with improvement of MoCA score ($R^2 = 0.54$, $p = 0.023$). Increased SUVRc in the frontal cortex and hippocampus was associated with unfavorable disability recovery (odds ratio [OR] = 7.1, 95% CI 1.67–29.9, $p = 0.008$) and persistent amnesia (OR = 5.4, 95% CI 1.3–22.2, $p = 0.021$) respectively.

Conclusion: Microglial activation visualized by [^{18}F]-DPA714 PET is associated with clinical features and may be used as a potential biomarker for therapeutic and prognostic evaluation.

The first two authors contributed equally to this work.

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1 | Introduction

Antibodies (Abs) against leucine-rich glioma-inactivated 1 (LGI1) protein could disturb synaptic transmission and lead to neuronal hyperexcitability [1]. Patients with anti-LGI1 encephalitis present severe cognitive impairment, refractory seizures, and psychological disorders [2]. The inflammatory reaction mediated by autoantibodies in the central nervous system (CNS), especially in the hippocampus and medial temporal lobe, is a hallmark of the pathogenesis for anti-LGI1-autoimmune encephalitis (AE) patients. According to a recent study, the sensitivity of traditional MRI scans is only 65% among all 134 patients and dramatically decreases to 45% for patients without autoantibodies in CSF, which might not be the most optimal method for diagnosis or therapeutic evaluation [2]. Earlier initiation of immunotherapy in the acute stage and sustained immunosuppressive medication is one of the most recommended etiology-oriented strategies for improving the prognosis of anti-LGI1 encephalitis [3, 4]. However, it is difficult to assess the efficacy of treatment and determine the discontinuation of long-term immunotherapy. Thus, more reliable technologies are urgently needed to evaluate the cerebral inflammatory level in LGI1-AE patients.

Positron emission tomography (PET) has been increasingly used in the diagnosis of AE as a promising technology allowing pathophysiological visualization without invasiveness [5]. [^{18}F]-fluorodeoxy-glucose PET ([^{18}F]-FDG-PET) showed higher sensitivity in detecting inflammatory lesions in AE [6]. Hypermetabolism in specific brain regions could be relevant to typical symptoms such as faciobrachial dystonic seizures (FBDS) in anti-LGI1 encephalitis [7]. Nevertheless, some unsolved issues about the FDG-PET application in AE-related research still exist. For instance, [^{18}F]-FDG PET abnormalities could not only be found in CNS inflammatory diseases but also in brain tumors, neurodegenerative, or psychological disorders [8, 9]. Since the cerebral glucose metabolism is influenced by different cell types, it is not clear to what extent the alteration of inflammation is reflected by [^{18}F]-FDG PET imaging changes [10, 11].

[^{18}F]-DPA714 is a selective radioligand binding with 18-kDa translocator protein (TSPO), expressed in microglia, astrocytes, as well as endothelial cells, but significantly upregulated in pro-inflammatory microglia in CNS inflammatory diseases [12]. Microglia are resident cells in humans brain and play vital roles in the adaptive immune response. Activated microglia could turn to a pro-inflammatory phenotype and secrete cytokines in response to pathogenic invasion as pathogen-associated molecular patterns (PAMPs) or homeostasis alteration as damage-associated molecular patterns (DAMPs) [13, 14]. Thus, [^{18}F]-DPA714 PET is recently considered a promising technology for monitoring cerebral inflammatory reactions during disease development based on visualizing microglial activation, especially for AE, which is difficult to determine by traditional imaging approaches.

Based on the Huashan Autoimmune encephalitis and Molecular Imaging Cohort (HS-ATOMIC), we firstly reported the application of [^{18}F]-DPA714 PET in anti-LGI1 encephalitis [15]. Here, we further investigated the correlation between regional TSPO uptake and clinical features as well as the microglia biomarker, soluble triggering receptor expressed on myeloid cells 2 (sTREM2) in CSF with a larger sample size. Additionally, we longitudinally studied the alterations of imaging patterns and neurological function recovery in patients with immunotherapy and evaluated the prognostic value of [^{18}F]-DPA714 PET based on clinical features with our follow-up cohort.

2 | Material and Methods

2.1 | Patients and Clinical Data

Patients with AE who received [^{18}F]-FDG PET scans or [^{18}F]-DPA714 PET scans in Huashan Hospital were enrolled in the Autoimmune encephalitis Molecular Imaging Cohort (HS-ATOMIC). Among them, patients diagnosed with anti-LGI1 encephalitis according to the criteria and confirmed independently by two senior neurologists were included in this study [16]. Clinical data at baseline was comprehensively collected, including results of physical examination, CSF examination, electroencephalograph (EEG), MRI, and tumor screening. Autoantibodies in patients' serum and CSF were detected by an in-house cell-based assay (CBA). A peripheral blood test for TSPO affinity genotyping was conducted as described before [17]. Disability and cognitive impairment levels were evaluated with the Modified Rankin scale (mRS) and Montreal cognitive assessment (MoCA) from baseline and reappraised every three months during the follow-up period. We defined unfavorable disability recovery as patients with an mRS score no less than 2 at the last time point during the follow-up. The acute stage of anti-LGI1 encephalitis was defined as the first three months from disease onset or worsening of previous symptoms, while others beyond that were defined as in the recovery stage [18, 19]. Age- and gender-matched patients with definite non-inflammatory diseases and healthy controls who underwent [^{18}F]-DPA714 PET scans were recruited in the non-inflammatory control (NIC) group. Written informed consent was obtained from all patients. The study protocol was approved by the institutional review board of Huashan Hospital, Fudan University.

2.2 | PET Imaging Acquisition and Imaging Analysis

According to previous studies on [^{18}F]-DPA714 kinetic modeling and application in neurological disorders, acquisition of PET images from 40 to 60 min after tracer injection was recommended [20, 21]. In our study, PET scanning for each patient was conducted 40 min after intravenous injection of 370 MBq of [^{18}F]-DPA714 with Siemens mCT flow edge 128 (Siemens, Germany) in 3D mode for 20 min and CT transmission scan for attenuation correction.

For imaging analysis, routine visual reading from three-dimensional aspects (transverse, sagittal, and coronal views) on Siemens Workstation was the first step to determine the TSPO uptake abnormalities compared with controls. Each case was reviewed by two experienced neuroradiologists independently, and the third senior radiologist would join the discussion to reach the consensus in case of disagreement. Then, computer-supported

reading was conducted using Statistical Parametric Mapping (SPM8) software (Wellcome Department of Imaging Neuroscience, Institute of Neurology, London, UK) implemented in MATLAB R2018b (Mathworks Inc., Sherborn, MA, USA). In SPM analysis, the preprocessing of scans from each subject was first spatially normalized into Montreal Neurological Institute (MNI) brain space and was then smoothed by a Gaussian filter of 10 mm FWHM to increase the signal to noise ratio. Next, we used radioactivity counts of the cerebellar cortex in each subject as a reference to obtain the individual intensity-normalized images. To characterize the neuroinflammation pattern in LGI1-AE patients compared with NIC, we performed a group comparison of PET images by using a two-sample t test according to the general linear model at each voxel. Moreover, to identify the areas displaying changes in TSPO retention underlying immunotherapy in longitudinal follow-up, we compared the baseline PET images with PET images after the treatment on a voxel-by-voxel basis using a 2-paired *t* test option. For both designs, we set the peak threshold at $p < 0.05$ to evaluate significant differences. Significant regions were localized by Talairach Daemon software (Research Imaging Center, University of Texas Health Science Center, San Antonio, TX, USA). The SPM maps for increased or decreased signal were overlaid on a standard T1-weighted MRI brain template in stereotaxic space.

Segmentation of regions of interest (ROI) was performed using the MRI imaging of individual patients with anti-LGI1 encephalitis in combination with an automated anatomical labeling atlas (AAL) with 70 brain regions. The mean standardized uptake values (SUVs) of [^{18}F]-DPA714 PET in brain regions from LGI1-AE patients were extracted. Since LGI1 has low expression in the cerebellum at both the protein and mRNA levels, and patients with anti-LGI1 encephalitis hardly present any cerebellar symptoms, the cerebellum could be a survivor when anti-LGI1 IgG attacks the CNS [22, 23]. Therefore, all obtained SUVs were subsequently normalized by the average SUV in the bilateral cerebellum, namely standardized uptake value ratios normalized to the cerebellum (SUVRc), which were used for further analysis.

2.3 | Determination of Microglial Activation Biomarker (sTREM2) With ELISA

CSF samples from LGI1-AE patients in the acute stage of disease were collected. Eighteen patients diagnosed with non-inflammatory diseases were included as controls, including cerebral venous sinus thrombosis ($n = 4$), idiopathic intracranial hypertension ($n = 4$), degenerative cervical spondylosis ($n = 3$), somatization disorder ($n = 3$), peripheral neuropathy ($n = 3$), and migraine ($n = 1$). All CSF samples were centrifuged at 1000×g for 10 min and stored at −80°C after lumbar puncture. Enzyme-linked immunosorbent assay (ELISA) kits were used to detect the expression level of sTREM2 following the manufacturer's instructions (ab224881, Abcam).

2.4 | Statistical Analysis

Demographics and clinical features were analyzed with the Tableone Package in R (version 4.2.3, R Foundation for Statistical Computing, Vienna, Austria). Continuous data were presented with median ± SD (IQR). Mann–Whitney *U* test was used to compare the difference in sTREM2 expression between anti-LGI1 encephalitis and controls. In semi-quantitative analysis for PET imaging, an unpaired Student *t* test with FDR correction was used for comparisons of SUVRc between subgroups. Spearman's analysis was performed to clarify the correlation between SUVRc and clinical scores including MoCA scale, IgG titers, Q-Alb, and sTREM2 expression. Statistical analyses were conducted using GraphPad Prism 10 (San Diego, CA, USA) and *p* values less than 0.05 were considered statistically significant.

3 | Results

3.1 | Clinical Characteristics of CSF LGI1 Antibody-Positive or Negative Patients that Underwent [^{18}F]-DPA714 PET Scan

Of the 122 AE patients with definite Abs from HS-ATOMIC, 46 patients (male, 66.7%) diagnosed as anti-LGI1 encephalitis with a median age of onset of 57 years (IQR: 48, 63) underwent [^{18}F]-DPA714 PET scan were enrolled in this study (Figure 1). Detailed clinical characteristics of patients at the time of the first PET scan were listed in Table 1 and S1. In terms of clinical disability and cognitive function, the median mRS score and MoCA score were 3 (IQR: 2, 3) and 22 (IQR: 17, 24.5), respectively. Antibodies against LGI1 were detected in all the patients (seropositivity: 95.6%, CSF positivity: 73.9%). The frequency of high-affinity-binding (HAB) genotype is 92.1% in all LGI1-AE patients. All patients with LGI1 encephalitis received immunotherapy. Thirty-one patients (69%) were in the acute stage with a median time of 27 days (IQR: 15, 48) from onset to PET scan, and all of them were receiving first-line immunotherapy (intravenous steroids, intravenous immunoglobulins, plasma exchange, or both) at the time of imaging. Fifteen patients were in the recovery stage when they underwent the PET scan, and they were also receiving long-term immunotherapy including oral steroids, mycophenolate mofetil, and Azathioprine. No significant difference was observed in symptomatic distribution or other clinical features including EEG, MRI scan, and CSF inflammatory indicators between subgroups in the acute and recovery stages.

Twenty-one NIC subjects consisted of 18 patients with neurodegenerative, psychiatric, and cerebrovascular diseases as well as 3 healthy volunteers. There was no significant difference in age (median 56 years, IQR: 41, 67), gender (male, 57.1%) and the proportion of HAB genotype between patients with LGI1-AE group and NIC group (92.1%). Detailed information on NIC was listed in Table S2.

3.2 | Correlation Between TSPO Distribution and Clinical Features in LGI1-AE Patients at Baseline

Voxel-based analysis was conducted with SPM software to compare TSPO imaging patterns between LGI1-AE patients and NIC subjects. Before that, comparison analysis of TSPO distribution between healthy volunteers and non-inflammatory patients was performed to validate the homogeneity in the NIC group. No significant difference was found between these two

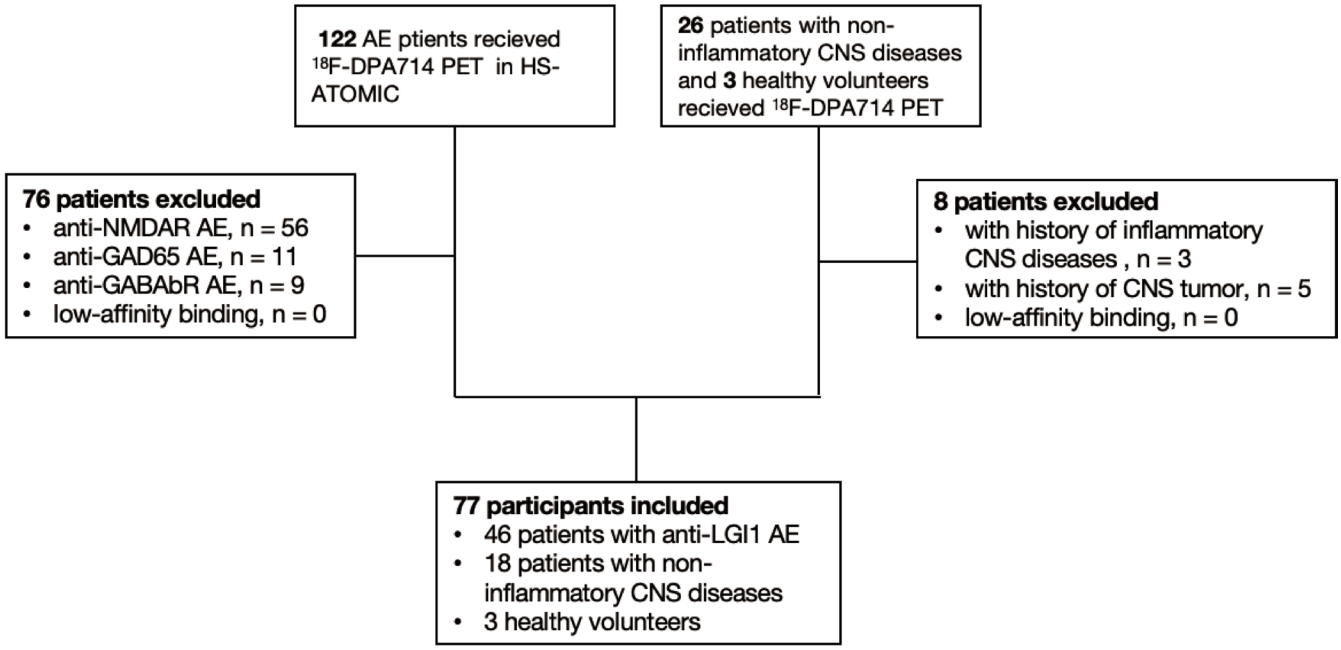


FIGURE 1 | Flow diagram of participants inclusion and exclusion.

TABLE 1 | Characteristics of patients with anti-LGI1 encephalitis at the time of PET scan.

Items	Overall LGI1 patients ($n=46$)	Patients in acute stage ($n=31$)	Patients in recovery stage ($n=15$)	p-value
Age at onset (years, median [IQR])	57 [48, 63]	56 [49.5, 62.5]	59 [39, 63]	0.879
Male, *n* (%)	30 (65.2)	21 (67.7)	9 (60)	0.852
Symptoms at the time of PET scan, *n* (%)				
Amnesia	41 (89.1)	28 (90.3)	13 (86.7)	> 0.99
Psychosis	27 (58.7)	17 (54.8)	10 (66.7)	0.533
Sleep disorder	19 (41.3)	12 (38.7)	7 (46.7)	0.752
Hyponatremia	18 (39.1)	14 (45.2)	4 (26.7)	0.336
FBDS	26 (56.5)	16 (51.6)	10 (66.7)	0.365
Autonomic disorders	21 (45.7)	15 (48.4)	6 (40)	0.754
Peripheral symptoms	9 (19.6)	6 (19.4)	3 (20)	> 0.99
Dizziness	5 (10.9)	5 (16.1)	0 (0)	0.157
mRS (median [IQR])	3.00 [2,3]	3.00 [2, 3]	3.00 [2, 3]	0.551
MoCA (median [IQR])	22.00 [17, 24.5]	22 [18.5, 24.5]	23 [14, 24.5]	> 0.99
MRI abnormalities in MTL, *n* (%)	27 (58.7)	19 (61.3)	8 (53.3)	0.752
Immunosuppressive treatment duration (days, median [IQR])	23 [10, 77.5]	12 [6.5, 23]	97 [79.5, 122]	< 0.001
Time from onset to PET imaging (days, median [IQR])	48 [21.5, 97.5]	27 [15, 48]	132 [100.5, 177]	< 0.001

Abbreviations: FBDS, faciobrachial dystonic seizures; LGI1, leucine-rich glioma-inactivated 1; MoCA, Montreal cognitive assessment; mRS, Modified Rankin scale (mRS).

groups across all brain regions ($p > 0.05$, Figure S1). In line with our previous study, patients with anti-LGI1 encephalitis had increased TSPO uptake in three typical regions, including the frontal cortex, hippocampus, and caudate nucleus (Figure 2C). Then, to semi-quantitatively analyze the association between TSPO distribution and cognitive impairment, correlation analysis was also conducted with SUVRc in these key regions above and MoCA score. Contemporaneous MoCA score was negatively

correlated with SUVRc in the hippocampus ($R^2 = 0.13$, $p = 0.034$) and frontal cortex ($R^2 = 0.13$, $p = 0.017$) of LGI1-AE patients, but not significantly correlated with SUVRc in the caudate nucleus (Figure 2D–F). We also investigated the influence of anti-LGI1 antibodies concentration and blood-brain barrier function on TSPO uptake. Neither anti-LGI1 IgG titer in CSF nor CSF/serum albumin quotient (Q-Alb) correlated with SUVRc in the key regions above ($p > 0.05$, Figure S2).

3.3 | Regional Increased TSPO Uptake Was Associated With Expression Levels of Microglial Activation Biomarker

The expression level of the microglial activation biomarker sTREM2 in CSF was determined in both the LGI1-AE group and the control group. As expected, patients with AE in the acute stage showed a higher concentration of sTREM2 in CSF compared to that of the control group ($p < 0.001$, Figure 3A), indicating more activated microglia in patients' brains. Besides, correlation analysis between the expression level of sTREM2 and the contemporaneous MoCA score suggested a significant negative association (Figure 3B). Interestingly, we also found that the CSF-sTREM2 concentration positively correlated with [^{18}F]-DPA714 PET SUVRc in the hippocampus ($R^2 = 0.18$, $p = 0.04$, Figure 3C). However, no significant correlation was found in other brain regions.

3.4 | Changes of [^{18}F]-DPA714 PET Imaging Patterns in Follow-Up of Anti-LGI1 Encephalitis

Of 46 patients with anti-LGI1 encephalitis, nine patients received baseline and second-time [^{18}F]-DPA714 PET scans after their immunotherapy for a median of 9 months (IQR: 6.5–12.5). All patients

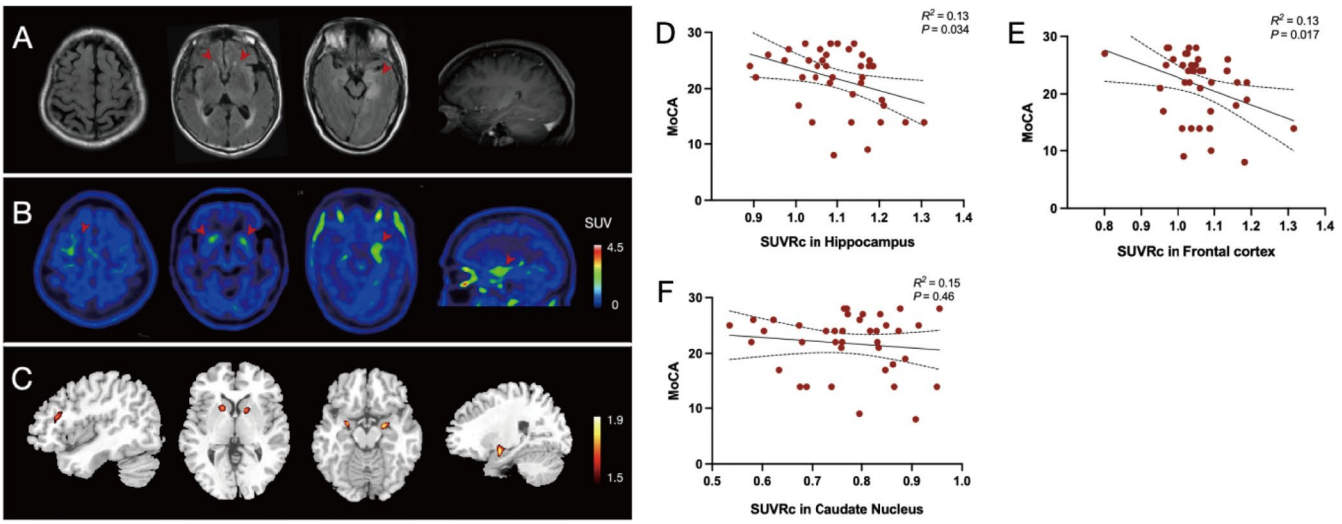


FIGURE 2 | Characteristics of TSPO-PET imaging for LGI1-AE patients and the correlation between regional TSPO distribution and cognitive impairment. (A) Typical T2-weighted fluid-attenuated inversion-recovery images with axial MRI scan showed obvious hyperintensity in left hippocampus and mild hyperintensity in bilateral caudate nucleus (red arrows). No obvious abnormality was observed in T1-weighted construct-enhanced image with sagittal MRI scan. (B) Typical ^{18}F-DPA714 PET images showed increased TSPO uptake in frontal cortex, caudate nucleus, and hippocampus (red arrows). (C) Voxel based comparison between patients with anti-LGI1 encephalitis and non-inflammatory controls (NIC). Increased SUVRc in LGI1-AE group was found in hippocampus, caudate nucleus, and frontal cortex. (D, E, F) Regional TSPO uptake was associated with cognitive impairment. MoCA score was negatively correlated with SUVRc in hippocampus ($R^2 = 0.13$, $p = 0.034$) and frontal cortex ($R^2 = 0.13$, $p = 0.017$), but not with caudate nucleus.

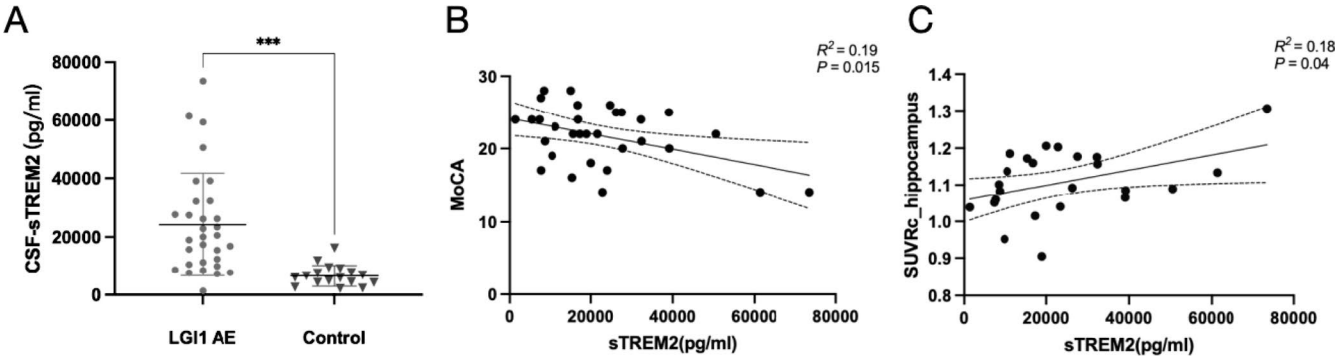


FIGURE 3 | Expression level of sTREM2 in CSF and its correlation with MoCA score and SUVRc in hippocampus. (A) Expression of sTREM2 in CSF was significantly increased in patients with anti-LGI1 encephalitis compared to controls (***$p < 0.005$). (B, C) Concentration of CSF sTREM2 in patients was negatively correlated with contemporary MoCA score ($R^2 = 0.19$, $p = 0.015$) and positively correlated with SUVRc in hippocampus ($R^2 = 0.18$, $p = 0.04$).

were in their recovery stage at the second PET scan without severe symptoms including epilepsy, amnesia, FBDS, and psychological disorders, except cognitive impairment. Improvement in daily activity and cognitive function reflected by decreased mRS scale and MoCA scale was also observed among those patients (Table S3). Among all brain regions, only TSPO uptake in the hippocampus showed a significant decrement after immunotherapy indicated by longitudinal PET scans (Figure 4A,B). The ratio of SUVRc decrement (ΔSUVRc ratio) in the hippocampus had a positive correlation with the ratio of MoCA increment (ΔMoCA ratio) (R^2 0.54, $p=0.023$, Figure 4C). However, we did not find the same changes in the frontal cortex and caudate nucleus.

3.5 | Multivariate Analysis for Association Between Regional SUVRc and Prognosis

To further discover the potential capacity of clinical outcome prediction by [^{18}F]-DPA714 PET, multivariate analysis was conducted with regional SUVRc and other risk factors based on univariate analysis and previous studies. Univariate analysis indicated that patients with unfavorable disability recovery (mRS ≥ 2) had higher SUVRc in the frontal cortex compared to those with favorable disability recovery (mRS <2, $p=0.002$, Figure 5A). Patients with persistent amnesia (MoCA <26) had higher SUVRc in the hippocampus compared to those with complete cognitive recovery (MoCA ≥ 26, $p=0.001$, Figure 5B). The age of disease onset, anti-LGI1 Abs positivity, mRS, and MoCA at baseline were taken into multivariate logistic regression analysis to construct a prediction model. Increased SUVRc in the frontal cortex predicted unfavorable disability recovery (odds ratio [OR] = 7.1, 95% CI 1.67–29.9, $p=0.008$). The final model predicted disability recovery with 83% sensitivity and 75% specificity with an AUC value of 0.84 (Figure 5C). Increased SUVRc in the hippocampus predicted persistent amnesia (OR = 5.4, 95% CI 1.3–22.2, $p=0.021$), while the MoCA score at baseline reduced the odds of persistent amnesia (OR = 0.8, 95% CI 0.6–0.9, $p=0.021$). The final model predicted persistent amnesia with 88% sensitivity and 81% specificity with an AUC value of 0.90 (Figure 5D).

4 | Discussion

Here, we characterized [^{18}F]-DPA714 PET imaging patterns and their correlation with neurological function and microglial activation in LGI1-AE patients by semi-quantitative analysis and longitudinal observation. Brain [^{18}F]-DPA714 PET was abnormal in 91.3% of patients with anti-LGI1 encephalitis, suggesting a higher sensitivity for detecting lesions compared to conventional MRI scan, EEG, and routine CSF test. Additionally, we also verified the previous findings; namely, apart from common abnormalities in the limbic system such as the hippocampus and medial temporal lobe, increased TSPO distribution also existed in widespread cortex and caudate nucleus. These alterations

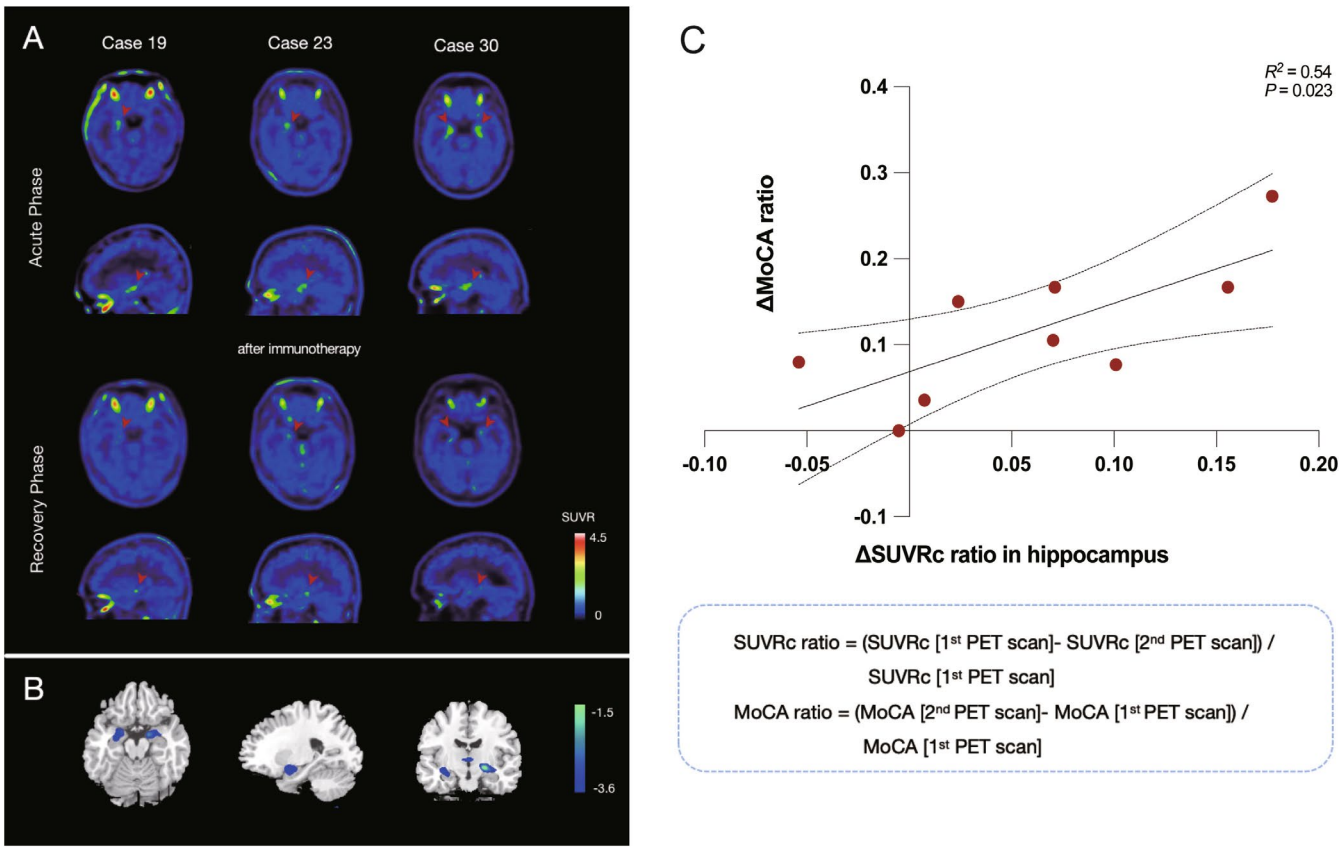


FIGURE 4 | Longitudinal TSPO-PET changes in LGI1-AE patients and its correlation with MoCA score. (A)Typical ^{18}F-DPA714 PET images of LGI1-AE patients with longitudinal scan, TSPO uptake in hippocampus (red arrows) decreased after immunotherapy. (B)Voxel based comparison between patients with anti-LGI1 encephalitis before and after treatment. (C) DSUVRc ratio in hippocampus was positively correlated with DMoCA ratio ($R^2=0.54$, $p=0.023$).

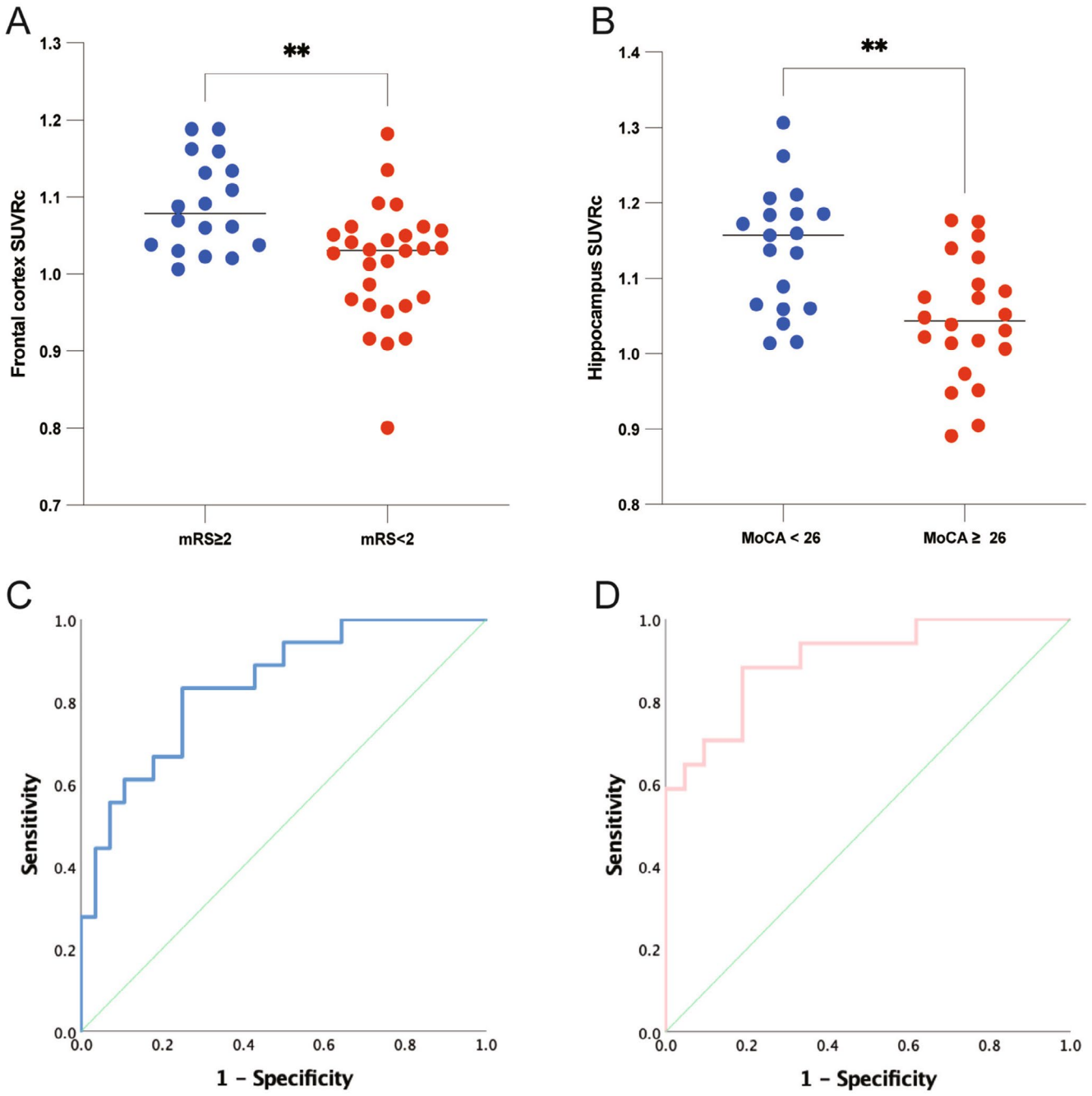


FIGURE 5 | Prognostic evaluation with SUVRc in hippocampus and frontal cortex. (A) Comparison of SUVRc in frontal cortex between patients with and without favorable disability recovery using univariate analysis. Patients with unfavorable disability recovery ($n = 18$) had higher SUVRc compared to those who had favorable recovery ($n = 28$) (B) Comparison of SUVRc in hippocampus between patients with different cognitive function using univariate analysis. Patients with persistent amnesia ($n = 19$) had higher SUVRc compared to those who had complete memory recovery ($n = 22$); $^{**}p < 0.01$. (C) Receiver operating characteristic curves (ROC) with AUC values based on the multivariate logistic regression analysis. SUVRc in frontal cortex had significant effect on predicting unfavorable disability recovery (AUC 0.84; sensitivity, 83%; specificity, 75%; $p < 0.001$). (D) Combination of SUVRc in hippocampus and MoCA score at baseline had significant effect on predicting sustained cognitive impairment (AUC 0.90; sensitivity, 88%; specificity, 81%; $p < 0.001$).

persisted in the recovery stage with a duration even exceeding 12 months, representing the long-term effect from activated microglia. For individual patients, TSPO uptake reduction has been observed in the hippocampus following immunotherapy, but not to the same extent in the cortex and caudate nucleus, which might be attributed to the higher level of microglial activation and more visible recovery in the hippocampus. Meanwhile, overexpression of sTREM2 was first evaluated in CSF of patients with anti-LGI1 encephalitis and positively correlated with SUVRc in the hippocampus, which provides another evidence for [^{18}F]-DPA714 PET as an imaging biomarker for microglial activation. A significant correlation was found between MoCA score and SUVRc in the hippocampus and frontal cortex. Additionally, we also found that SUVRc in the hippocampus

and frontal cortex at baseline has potential prognostic capability. On the other hand, our longitudinal study also found that MoCA score improvement was associated with SUVRc reduction in the hippocampus. All these results suggested that [^{18}F]-DPA714 PET is not only the sign of inflammatory reaction but also a potential biomarker for disease severity evaluation.

PET imaging labeled on TSPO protein has been applied in clinical practice for decades. Recently, with the development of a higher-affinity radiotracer, it is becoming a hot topic for diagnosing neurological disorders as a hallmark of inflammatory reaction [24, 25]. Based on our analysis, [^{18}F]-DPA714 PET also had higher sensitivity to detect inflammatory regions before the cerebral structural damage in patients with anti-LGI1 encephalitis compared to conventional MRI scans. Anti-LGI1 encephalitis was traditionally considered an encephalitis that only involved the limbic lobe, but recent evidence has also indicated that the biological function of the frontal cortex could also be influenced by autoantibodies. Apart from hypermetabolic changes in the hippocampus, asymmetric cortical hypometabolism was also observed in LGI1-AE patients with [^{18}F]-FDG PET, which suggested a potential underlying cause for the persistent amnesia and seizures [26, 27]. Meanwhile, based on functional MRI imaging research, a large-scale brain network disruption beyond apparent damage in the hippocampus was revealed in anti-LGI1 encephalitis, representing a compensatory mechanism for cognitive impairment [28]. Thus, the application of [^{18}F]-DPA714 PET could pave a new way for expanding our understanding of whole-cerebral disruption in limbic encephalitis. Additionally, we also found that a higher TSPO uptake level in the hippocampus and frontal cortex was associated with more severe cognitive impairment, which has not been confirmed by our previous research [17].

Patients with anti-LGI1 encephalitis receiving immunotherapy usually had a favorable outcomes with seizure-free status and neurological function improvement [29]. Nevertheless, most of them remained with moderate or mild cognitive decline, and 13%–35% of patients experienced relapses during the follow-up [3, 29–31]. One of the longitudinal studies based on multimodal MRI found global cerebral atrophy in LGI1-AE patients and a correlation between cognitive function and the volume of gray matter as well as putamen [32]. Another study observed hypermetabolism in the mediotemporal lobe and putamen detected by [^{18}F]-FDG PET, which could be recovered after immunotherapy [33]. However, longitudinal studies on inflammatory changes after the immunotherapy of anti-LGI1 encephalitis were lacking in existing studies. Meanwhile, although there has been an increased application of second-line immunotherapy, there is little consensus among clinicians on when these treatments should be withdrawn due to a lack of objective evidence [34]. Here, we first elucidate the inflammatory alterations in LGI1-AE patients with longitudinal brain [^{18}F]-DPA714 PET scans and semi-quantitative analysis. Patients in the acute stage had higher SUVRc in the hippocampus compared to that in the recovery stage. The decreasing SUVRc in the hippocampus mirrors the improvement of cognitive function, suggesting that [^{18}F]-DPA714 PET could be a potential marker to evaluate cerebral inflammatory changes during treatment. However, it should be emphasized that, since the roles of microglia in LGI1-AE remain incompletely understood, a decrease in TSPO uptake should not be considered sufficient justification for withdrawing immunotherapy in clinical practice.

In addition to visualizing inflammation changes, [^{18}F]-DPA714 PET is also considered a promising technique for immunotherapeutic evaluation and to predict disability accumulation in various neuroimmune diseases, especially MS, which has been validated with a series of cohort studies [35–37]. Previous longitudinal studies revealed that hypometabolism in limbic cortical regions at baseline could be associated with higher ultimate mRS scores during long-term follow-up [33]. However, the evaluation of prognostic effect in AE with this novel technique has not been investigated yet. In our study, we firstly analyzed the relationship between regional TSPO uptake at baseline and prognosis with a multiple regression model by controlling possible interference parameters, illustrating that higher SUVRc in the hippocampus and frontal cortex could be risk factors for persistent amnesia and unfavorable disability recovery, respectively. Therefore, apart from regional metabolic features, primary neuroinflammatory patterns at the onset of encephalitis visualized by [^{18}F]-DPA714 PET could be another marker for predicting clinical outcomes for anti-LGI1 encephalitis.

To the best of our knowledge, microglial cells are activated when confronting pathogen-associated molecular patterns (PAMPs) and damage-associated molecular patterns and perform both pro-inflammatory and anti-inflammatory functions by altering their morphology and gene expression in neuroinflammatory disorders [38]. TSPO protein is mainly expressed in microglia, astrocytes, and epithelial cells, but an increasing number of in vivo studies with mouse models showed that the proliferation of Iba1+ microglia is a predominant contributor during the process of neuroinflammation [12, 39]. Microglial activation was observed and performs numerous functions in various neuroimmune disorders. For instance, activated microglia were found in normal appearing white matter of MS patients' post-mortem brain tissue and expressed pro-inflammatory cytokines like CCL2 in the animal models of demyelination [40, 41]. However, the role of activated microglia in AE has not been fully investigated. Recent studies suggested that activated microglia were co-localized with NMDAR-IgG complexes and exerted a phagocytic effect with IgG-bound receptors, which might be another contributor to the reduction of neuronal NMDARs apart from direct binding with autoantibodies [42]. Therefore, further investigating the contribution of activated microglia will provide a novel insight into understanding pathological changes in AE.

In our study, we intended to reveal the relationship between [^{18}F]-DPA714 PET imaging and microglial activation by comparing the correlation between regional TSPO uptake and the expression of sTREM2, a biomarker of activated microglia in human CSF. Overexpression of sTREM2 has been reported in the CSF of patients with neurodegenerative and neuroinflammatory disorders including Alzheimer's disease (AD), Parkinson disease, MS, and anti-NMDAR encephalitis [43–46]. Here, we also observed that the CSF expression level of sTREM2 in LGI1-AE patients was increased and positively correlated with cognitive impairment, which is in line with previous studies on AD and anti-NMDAR AE [46, 47]. Moreover, we firstly discovered the association between [^{18}F]-DPA714 PET imaging and sTREM2 expression in AE, reinforcing the theory that increasing TSPO uptake represents microglial activation.

There are some limitations in our study. Firstly, this study was based on a single-center cohort with a relatively short follow-up duration; the prognostic and monitoring capabilities of [^{18}F]-DPA714 PET imaging could not be accurate. Secondly, given a variety of immunotherapies, the treatment that patients received could suppress inflammation differently and lead to heterogeneity of PET imaging patterns even among patients in the same disease phase. Further studies with a larger sample size could focus on particular subgroups based on immunotherapeutic strategy. Thirdly, the sample size of patients with longitudinal PET scans was limited, and the intervals between two scans were short. Thus, apart from the hippocampus, the recovery of TSPO uptake in other brain regions was likely underestimated. Lastly, sTREM2 was the only CSF biomarker included in our study, which is unable to comprehensively reflect the inflammatory level in the CNS. Other biomarkers or downstream cytokines related to microglial activation, such as tumor necrosis factor-α, interleukin-18, and interleukin-1β could be taken into consideration in future research for characterizing the pro-inflammatory function of activated microglia.

5 | Conclusion

This study demonstrated that increased TSPO uptake in the hippocampus, caudate nucleus, and frontal cortex was a hallmark for patients with anti-LGI1 encephalitis. Regional SUVRc was associated with cognitive impairment and microglial activation. Immunosuppressive therapy decreased hippocampal TSPO uptake along with memory recovery. Our study also emphasized the potential roles of baseline [^{18}F]-DPA714 PET in prognostic evaluation, which provided a novel biomarker for individualized treatment of anti-LGI1 encephalitis. More research with a multicenter cohort and longer follow-up duration would be warranted to further validate the application of [^{18}F]-DPA714 PET in AE.

Author Contributions

Jingguo Wang: conceptualization, methodology, data curation, formal analysis, writing – review and editing, investigation. **Jingjie Ge:** conceptualization, investigation, methodology, data curation, writing – original draft. **Bo Deng:** data curation, resources, investigation, writing – review and editing. **Huamei Lin:** investigation, data curation, resources. **Wenbo Yang:** data curation, resources, investigation. **Tianyang Sheng:** investigation, data curation, resources. **Weijun Tang:** data curation, formal analysis, resources. **Hai Yu:** data curation, formal analysis, resources. **Xiang Zhang:** data curation, writing – review and editing, investigation. **Yarong Li:** data curation, resources. **Xiaoni Liu:** data curation, resources. **Chuantao Zuo:** conceptualization, methodology, project administration, writing – original draft, data curation. **Xiangjun Chen:** conceptualization, methodology, project administration, writing – original draft, funding acquisition.

Ethics Statement

This study was approved by the Ethics Committee of Huashan Hospital, Fudan University (KY2021-540).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data are available from the corresponding author upon reasonable request.

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